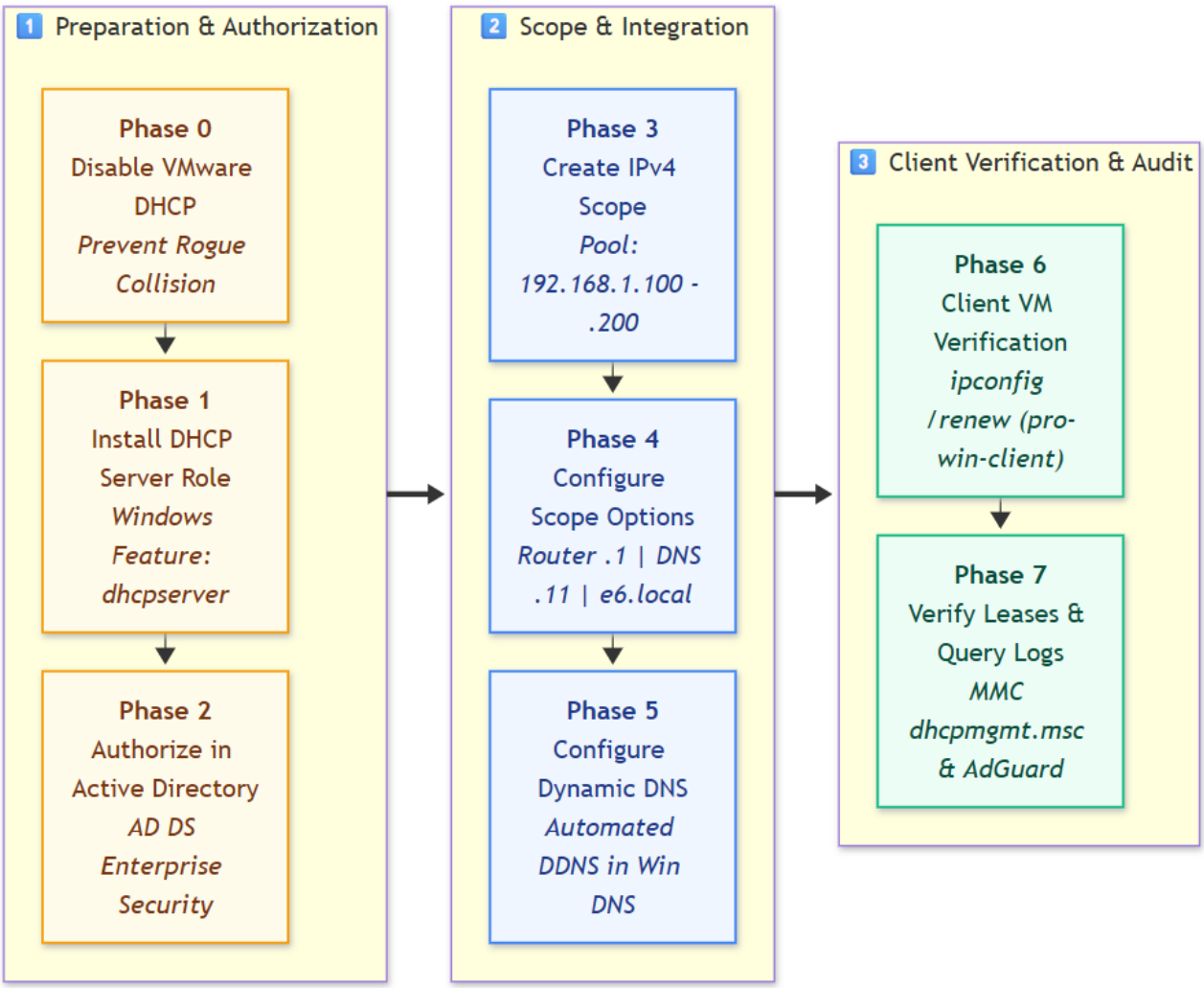
TECHNICAL SPECIFICATION & IMPLEMENTATION GUIDE

Enterprise DHCP Server Implementation Guide

Overview & Workflow Roadmap

This implementation guide provides complete, battle-tested step-by-step instructions (both Graphical User Interface and PowerShell) to deploy an enterprise-grade Windows DHCP Server integrated with our Two-Tier DNS Architecture (`192.168.1.10` Windows DNS + `192.168.1.11` AdGuard Home Docker).



Master Subnet & IP Pool Reference (`192.168.1.0/24`)

Network Parameter	Value	Description
Subnet ID	`192.168.1.0`	Class C private network
Subnet Mask	`255.255.255.0` (`/24`)	256 total IP addresses
Default Gateway	`192.168.1.1`	VMware Virtual NAT Router (`vmnat.exe`)
Windows Server (AD DC)	`192.168.1.10`	Domain Controller, Windows DNS, DHCP Server
AdGuard Home (Docker)	`192.168.1.11`	Network-wide ad blocker & Cloudflare DoH
Static Server Pool	`192.168.1.12` - `192.168.1.99`	Excluded from DHCP (infrastructure servers)
DHCP Distribution Pool	**`192.168.1.100` - `192.168.1.200`**	**Dynamic IP leasing range for client VMs**
Scope Option 003 (Router)	`192.168.1.1`	Assigned to all DHCP clients
Scope Option 006 (DNS)	**`192.168.1.11`**	**Assigned to all DHCP clients (AdGuard)**
Scope Option 015 (Domain)	`e6.local`	Search suffix for Active Directory
Lease Duration	`8 days` (Default)	Standard enterprise lease

Phase 0: Disable VMware Workstation Built-in DHCP (Critical!)

> 🛑 **CAUTION —**
>
> **Why this step is mandatory:**

> VMware Workstation includes its own virtual DHCP server (`vmnetdhcp.exe`) on `VMnet8`. If you run Windows Server DHCP while VMware's DHCP is active, a **DHCP Race Condition** occurs. Whichever server replies first gives the client its IP. If VMware wins, the client receives VMware's DNS (`192.168.1.2`), completely bypassing AdGuard and breaking Active Directory domain resolution!

Execution Steps (On Host Laptop: Windows 11):

1. On your physical host machine (Windows 11), open the Start Menu and type:

Virtual Network Editor (Run as Administrator).

2. Click **Change Settings** at the bottom-right (shield icon) to grant Administrator privileges.
3. In the top table, select **`VMnet8` (NAT)**.
4. At the bottom, look for the checkbox:

❌ **Uncheck: "Use local DHCP service to distribute IP address to VMs"**.

5. Click **Apply**, then click **OK**.

(Alternative via Windows 11 Host Services: Open `services.msc` ➔ locate VMware DHCP Service ➔ Right-click ➔ Stop, and set Startup type to Disabled).

Phase 1: Install DHCP Server Role on Windows Server 2022

Perform these steps inside your **Windows Server VM (`pro-win-server`)**:

Option A: Via PowerShell (Fastest)

Open PowerShell as Administrator:

```
# Install the DHCP server role and RSAT management tools
Install-WindowsFeature -Name DHCP -IncludeManagementTools

# Verify installation status
Get-WindowsFeature -Name DHCP
```

Option B: Via Server Manager GUI

6. Open **Server Manager** ➔ Click **Manage** (top-right) ➔ Select **Add Roles and Features**.
7. Click **Next** until you reach the **Server Roles** page.
8. Check the box for **DHCP Server**.
9. In the popup window (*Add features that are required for DHCP Server?*), click **Add Features**.
10. Click **Next ➔ Next ➔ Next ➔** Click **Install**.
11. When the installation finishes, click **Close**.

Phase 2: Authorize DHCP Server in Active Directory (AD DS)

> 📌 **NOTE —**
>
> In an Active Directory forest, an unauthorized DHCP server is treated as a rogue device and will refuse to distribute IP leases. You must authorize the DHCP server in Active Directory.

Option A: Via PowerShell

```
# Add DHCP security groups
netsh dhcp add securitygroups

# Restart the DHCP service to apply security group membership
Restart-Service dhcpserver

# Authorize this server in Active Directory forest (e6.local)
Add-DhcpServerInDC -DnsName WIN-J17IMHCEMA9.e6.local -IPAddress 192.168.1.10

# Verify authorization
Get-DhcpServerInDC
```

Option B: Via Server Manager GUI

12. In **Server Manager**, look at the top-right notification flag (yellow exclamation mark ⚠).
13. Click the notification ➔ Click "**Complete DHCP configuration**".

14. In the DHCP Post-Install configuration wizard:

- **Description page:** Click **Next**.
- **Authorization page:** Select **"Use the following user's credentials"** (should show `E6\Administrator`) ➔ Click **Commit**.

15. Click **Close**.

Phase 3: Create IPv4 DHCP Scope (`192.168.1.0/24`)

We will create a scope named **`E6-Clients-Scope`** distributing IPs from `192.168.1.100` to `192.168.1.200`.

Option A: Via PowerShell

```
# Create the DHCP Scope
Add-DhcpServerv4Scope -Name "E6-Clients-Scope" `
    -StartRange 192.168.1.100 `
    -EndRange 192.168.1.200 `
    -SubnetMask 255.255.255.0 `
    -State Active `
    -LeaseDuration (New-TimeSpan -Days 8) `
    -Description "Scope for lab client virtual machines"
```

Option B: Via DHCP Management Console GUI (`dhcpmgmt.msc`)

16. Press `Win + R` ➔ type **`dhcpmgmt.msc`** ➔ press **Enter**.
17. In the left tree, expand your server name: **`WIN-J17IMHCEMA9.e6.local`**.
18. Right-click **IPv4** ➔ Select **New Scope...**.
19. Click **Next** on the welcome screen.
20. **Scope Name:**

- **Name:** `E6-Clients-Scope`
- **Description:** `Scope for lab client virtual machines`
- Click **Next**.

21. **IP Address Range:**

- **Start IP address:** `192.168.1.100`
- **End IP address:** `192.168.1.200`
- **Length:** `24`
- **Subnet mask:** `255.255.255.0`
- Click **Next**.

22. **Add Exclusions and Delay:**

- Leave blank (our range already starts at `.100`, excluding our `.1` to `.99` static infrastructure) ➔ Click **Next**.

23. **Lease Duration:**

- Leave default (`8 Days`) ➔ Click **Next**.

24. **Configure DHCP Options:**

- Select **"Yes, I want to configure these options now"** ➔ Click **Next**.

Phase 4: Configure Scope Options (Router, DNS, Domain)

This is the most critical configuration phase that connects DHCP to AdGuard Home.

1. Scope Option Settings Table

Option Code	Parameter Name	Setting Value	Purpose
`003`	**Router (Default Gateway)**	`192.168.1.1`	VMware Virtual NAT Gateway
`006`	**DNS Servers**	**`192.168.1.11`**	**AdGuard Home Container (Port 53)**
`015`	**Domain Name**	`e6.local`	Primary Active Directory DNS Suffix

Option A: Via PowerShell

```
# Set Router (Gateway) -> 192.168.1.1
Set-DhcpServerv4OptionValue -ScopeId 192.168.1.0 -OptionId 3 -Value "192.168.1.1"

# Set DNS Server -> 192.168.1.11 (AdGuard Home)
Set-DhcpServerv4OptionValue -ScopeId 192.168.1.0 -OptionId 6 -Value "192.168.1.11"

# Set Domain Name (DNS Suffix) -> e6.local
Set-DhcpServerv4OptionValue -ScopeId 192.168.1.0 -OptionId 15 -Value "e6.local"
```

Option B: Via GUI (Continuation of New Scope Wizard)

25. **Router (Default Gateway) [Option 003]:**

- In IP address box, type: **`192.168.1.1`** ➔ Click **Add** ➔ Click **Next**.

26. **Domain Name and DNS Servers [Option 006 & 015]:**

- **Parent domain:** Type **`e6.local`**.
- Under IP address:
- Remove any existing addresses (like `127.0.0.1` or `192.168.1.10`).
- Type: **`192.168.1.11`** ➔ Click **Add**.

(If a prompt says "The DNS server is not valid or does not respond", click Yes to add it anyway).

- Click **Next**.

27. **WINS Servers:**

- Leave blank ➔ Click **Next**.

28. **Activate Scope:**

- Select **"Yes, I want to activate this scope now"** ➔ Click **Next** ➔ Click **Finish**.

Phase 5: Configure Dynamic DNS (DDNS) Integration

Configure DHCP to automatically register client lease hostnames into Windows Active Directory DNS (`e6.local`).

Option A: Via PowerShell

```
# Enable Dynamic DNS updates for clients in the scope
Set-DhcpServerv4DnsSetting -ScopeId 192.168.1.0 `
    -DynamicDnsUpdates "Always" `
    -DeleteDnsRROnLeaseExpiry $true `
    -UpdateDnsRRForClientsNotRequestingUpdates $true
```

Option B: Via GUI (`dhcpmgmt.msc`)

29. In **DHCP Manager**, expand **IPv4 ➔** Right-click **Scope [192.168.1.0] E6-Clients-Scope ➔** Click **Properties**.
30. Select the **DNS** tab:

- Check: **"Enable DNS dynamic updates according to the settings below:"**
- Select: **"Always dynamically update DNS records"**.
- Check: **"Discard A and PTR records when lease is deleted"**.
- Check: **"Dynamically update DNS A and PTR records for DHCP clients that do not request updates"**.

31. Click **Apply ➔** Click **OK**.

Phase 6: Client VM Verification (`pro-win-client`)

Now test the complete DORA cycle by configuring your client VM (**`pro-win-client`**) to obtain its IP address automatically.

Step 1: Set Client to DHCP Mode

On **`pro-win-client`**:

Via GUI (`ncpa.cpl`):

32. Press `Win + R` **➔** type **`ncpa.cpl`** **➔** press **Enter**.
33. Right-click **Ethernet0 ➔** select **Properties**.
34. Double-click **Internet Protocol Version 4 (TCP/IPv4)**:

- Select: ◯ **"Obtain an IP address automatically"**
- Select: ◯ **"Obtain DNS server address automatically"**

35. Click **OK** ➔ Click **Close**.

Via PowerShell (Alternative):

```
Set-NetIPInterface -InterfaceAlias "Ethernet0" -Dhcp Enabled
Set-DnsClientServerAddress -InterfaceAlias "Ethernet0" -ResetServerAddresses
```

Step 2: Trigger the DORA Exchange

In PowerShell on **`pro-win-client`**:

```
# Release current IP
ipconfig /release

# Request a new lease from Windows DHCP (Triggers DORA!)
ipconfig /renew

# Inspect all received parameters
ipconfig /all
```

Step 3: Verify Received Configuration

Examine the `ipconfig /all` output on the client. It must show:

```
Ethernet adapter Ethernet0:
   Connection-specific DNS Suffix  . : e6.local
   Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv4 Address. . . . . . . . . . . : 192.168.1.100(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Friday, October 9, 2026
   Lease Expires . . . . . . . . . . : Saturday, October 17, 2026
   Default Gateway . . . . . . . . . : 192.168.1.1
   DHCP Server . . . . . . . . . . . : 192.168.1.10
   DNS Servers . . . . . . . . . . . : 192.168.1.11
```

- ☑ **DHCP Server:** Shows `192.168.1.10` (Windows Server issued the lease!).
- ☑ **IPv4 Address:** Shows `192.168.1.100` (First address in our pool).
- ☑ **DNS Server:** Shows `192.168.1.11` (AdGuard Home).
- ☑ **Gateway:** Shows `192.168.1.1` (VMware NAT router).

Step 4: Run End-to-End DNS Verification Suite on Client

```
# 1. Test Active Directory Resolution
nslookup WIN-J17IMHCEMA9.e6.local

# 2. Test Short Computer Name (Option 015 Suffix Resolution)
ping WIN-J17IMHCEMA9

# 3. Test Network-Wide Ad Blocking
nslookup adservice.google.com

# 4. Test Encrypted Internet Browsing
nslookup google.com
```

Phase 7: Verification in DHCP MMC & AdGuard Query Log

1. View Active Leases in Windows Server DHCP MMC

On **Windows Server**:

36. Press `Win + R` ➔ type **`dhcpmgmt.msc`** ➔ press **Enter**.
37. Expand: **`WIN-J17IMHCEMA9.e6.local`** ➔ **`IPv4`** ➔ **`Scope [192.168.1.0] E6-Clients-Scope`** ➔ Click **Address Leases**.
38. In the right pane, you will see your client VM listed:

- **Client IP Address:** `192.168.1.100`
- **Host Name:** `pro-win-client.e6.local`
- **Lease Expiration:** 8 days from today
- **Type:** DHCP

Via PowerShell:

```
Get-DhcpServerv4Lease -ScopeId 192.168.1.0
```

2. Verify Dynamic DNS Registration in Windows DNS MMC

39. Press `Win + R` ➔ type **`dnsmgmt.msc`** ➔ press **Enter**.
40. Expand **Forward Lookup Zones** ➔ Click **`e6.local`**.
41. Notice that a new `A` record for **`pro-win-client`** pointing to **`192.168.1.100`** was created automatically by DHCP!

3. Verify in AdGuard Dashboard

42. Open `http://localhost:8080/#querylog` on the server.
43. Watch incoming DNS requests from client **`192.168.1.100`**!

Phase 8: Troubleshooting & Diagnostic Reference

1. Client Receives IP from Wrong Range (e.g. `192.168.1.128+`)

- **Root Cause:** VMware's built-in DHCP (`vmnetdhcp.exe`) is still running on `VMnet8`.
- **Fix:** Open **Virtual Network Editor** on Windows 11 host ➔ Uncheck *"Use local DHCP service to distribute IP address to VMs"* on `VMnet8` ➔ Click Apply. On client, run `ipconfig /release && ipconfig /renew`.

2. DHCP Server Icon Has Red Exclamation / Refuses Leases

- **Root Cause:** DHCP server is not authorized in Active Directory.
- **Fix:** In PowerShell on the server, run:

```
Add-DhcpServerInDC -DnsName WIN-J17IMHCEMA9.e6.local -IPAddress 192.168.1.10
Restart-Service dhcpserver
```

3. Client Receives IP but Cannot Access Internet

- **Root Cause:** Option 003 (Router) is misconfigured or pointing to an invalid gateway.
- **Fix:** Verify Scope Option 003 is set to `192.168.1.1` (VMware Virtual NAT Gateway). Test with `ping 192.168.1.1` from the client.